# Process Book: Project Number 8

**Scouting the Ultimate Vertical Playmaker from the 2022 World Cup**

## 1. The Initial Concept & The Pivot

Initially, my idea for this project was completely different. I wanted to analyze the recent resurgence of "Y2K vintage digital cameras." My plan was to build an eBay scraper using Python to pull pricing data, bid frequencies, and model popularity to visualize the hype cycle of different digicam brands. I then realized that eBay only keeps purchased history for items up to 90 days, so I cannot scrape info for years before the trend, at the peak, and after the trend. This made my initial plan worthless.

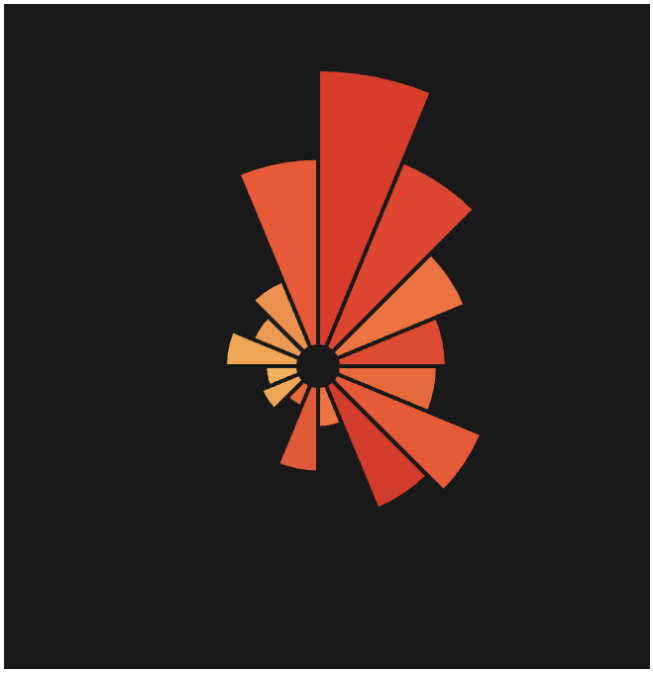
However, during my research phase, I discovered **StatsBomb**, an industry-leading football analytics company that released their entire 2022 Men's World Cup dataset for free. As an avid player of FIFA Career Mode, where scouting and analyzing for hidden gems is my favorite part of the game, I immediately pivoted. I decided to treat this project as a real-world scouting assignment.

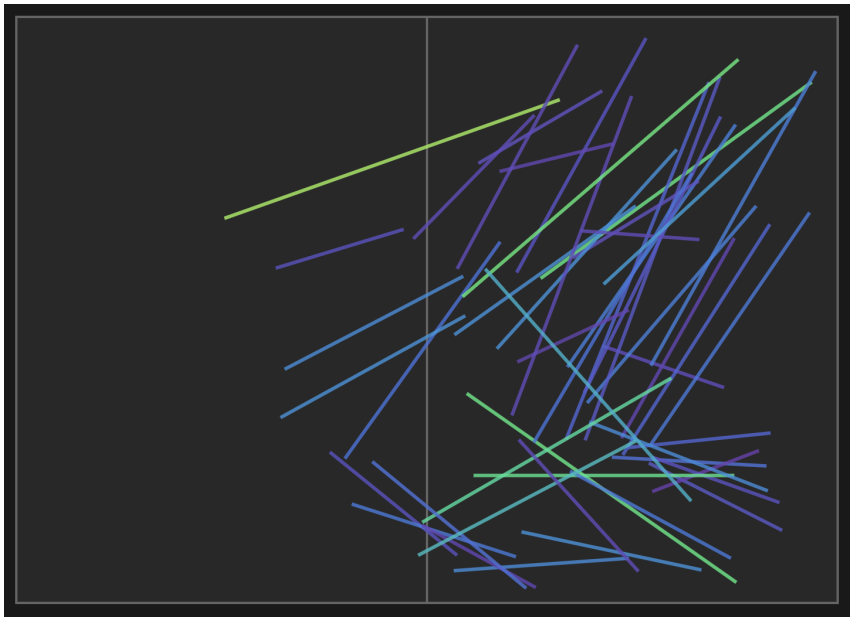
**The Premise:** In real life, massive clubs use the World Cup as a scouting ground. They often watch players' performances in this biggest tournament and try to buy the players after the World Cup ends. My goal was to build a data-driven dossier to find the ultimate "Number 8" (a playmaking central midfielder) from the 2022 tournament.

## 2. Inspiration & Sketches

A few weeks ago, during a class reflection, I analyzed a specific type of data visualization called a **Pass Sonar**. Pass sonars are brilliant because they don't just show *where* a player passes, but *how* they think.

I knew I wanted to use this technique as the climax of my project. I began sketching out ideas of how a Pass Sonar could be paired with a vertical pitch map to show both the "Idea" (the sonar) and the "Execution" (the pitch map).



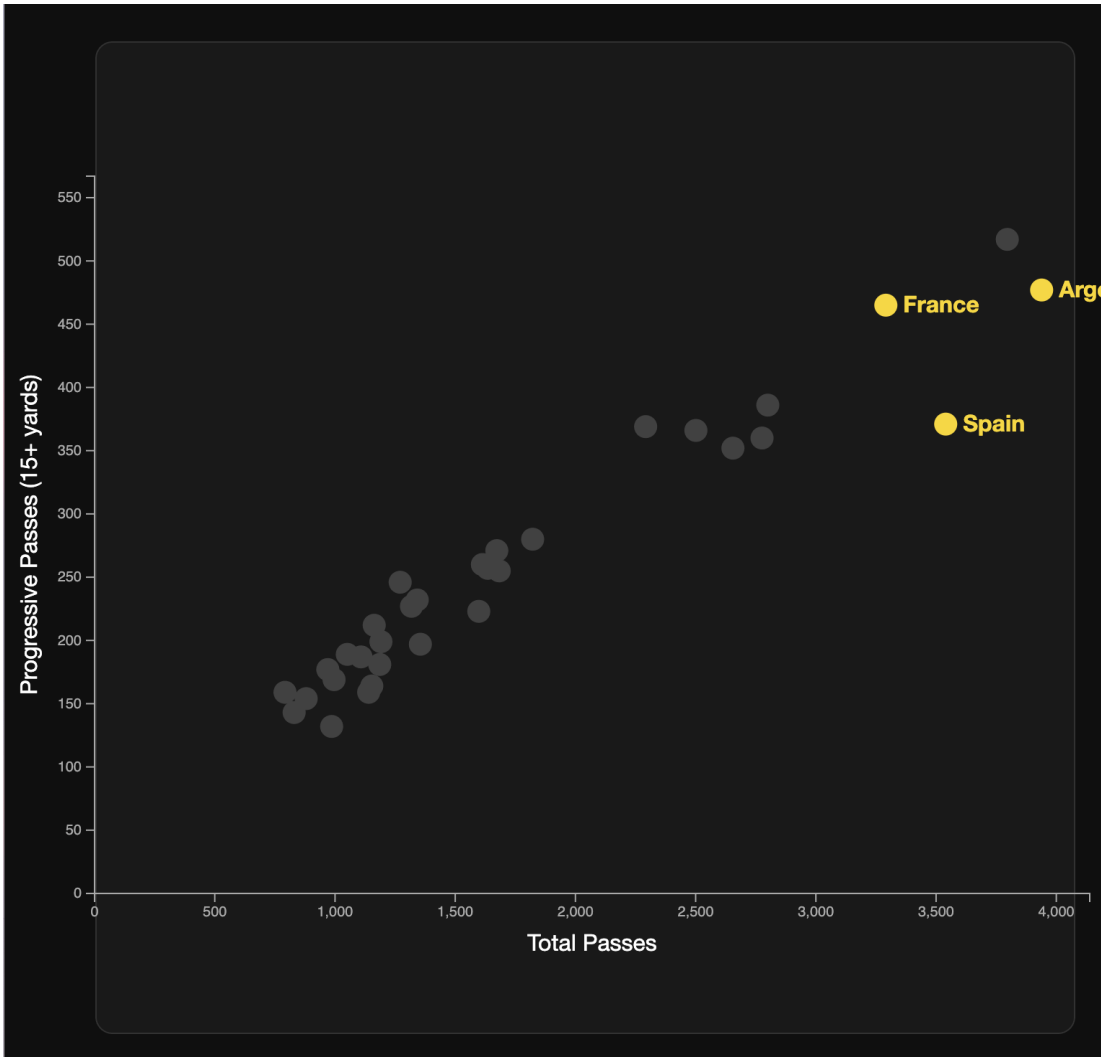


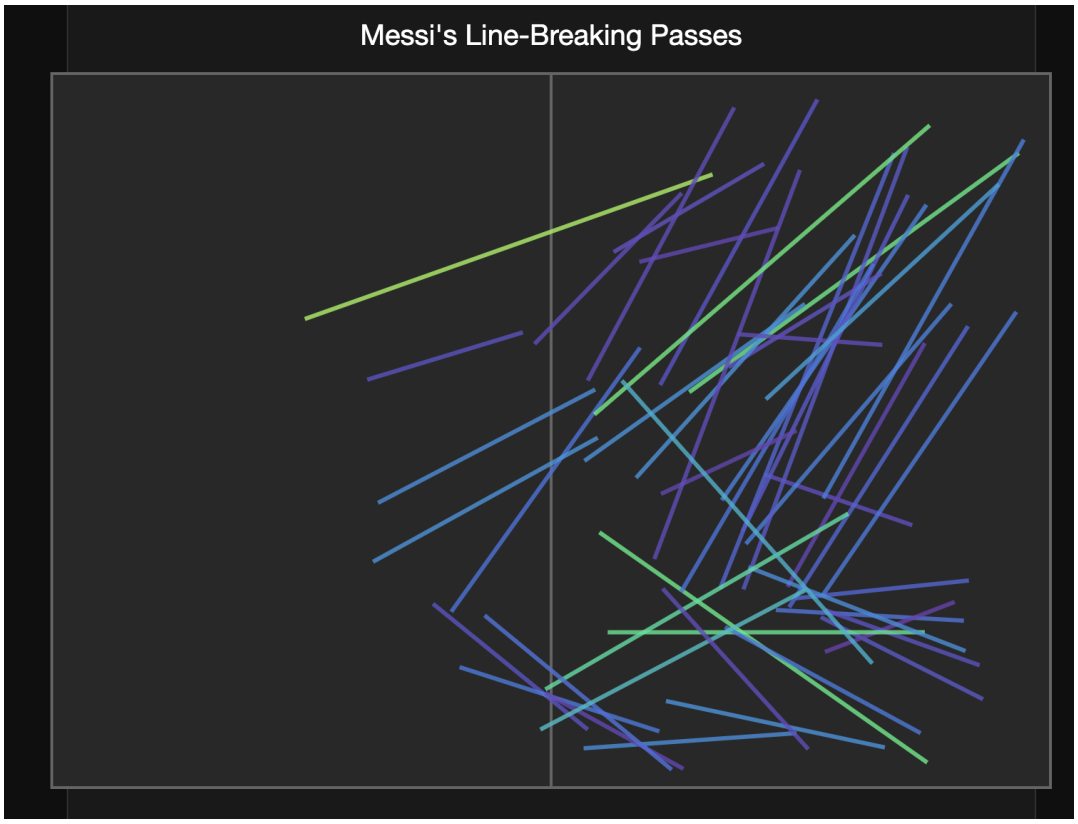
# 3. The Evolution of the Visualization: Trial and Error

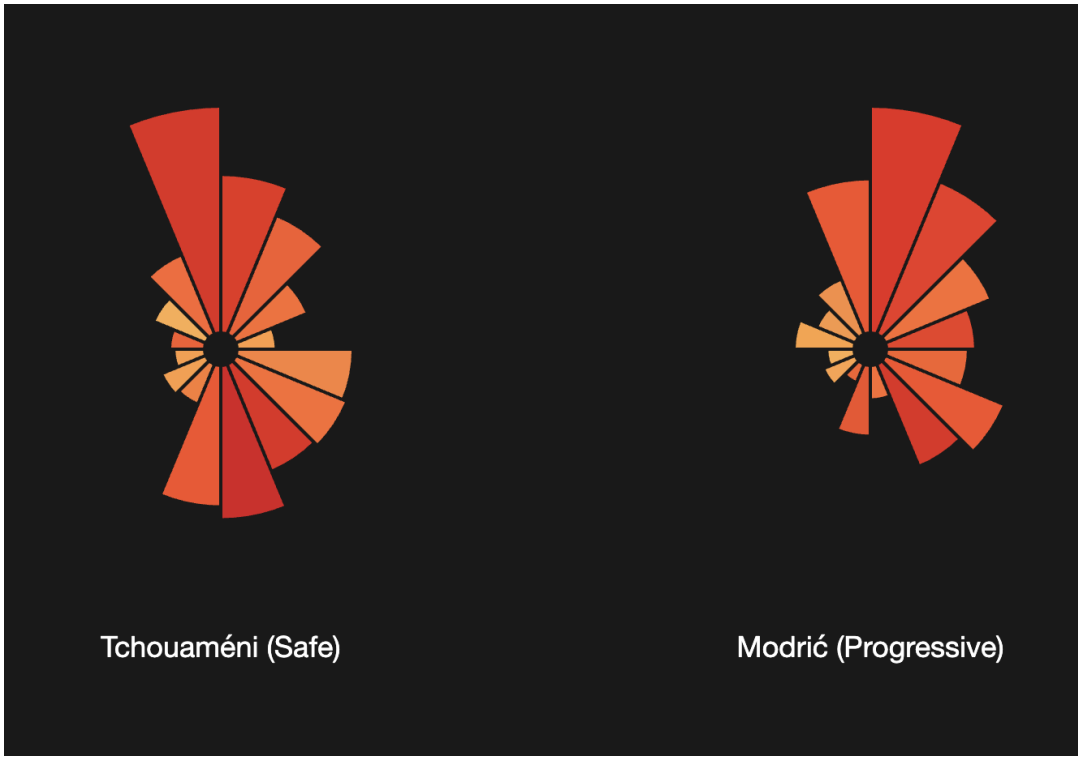
Before arriving at the final "Scouting Funnel," I went through three distinct iterations. I started with a pre-written narrative in my head called "The Illusion of Possession," focusing on how Spain passed a lot but failed, while Argentina and France succeeded. Building these early drafts taught me crucial lessons about data normalization and narrative bias.

## Attempt 1: The Absolute Totals Trap

- **The Concept:** I wanted to show that completing a thousand sideways passes doesn't win matches.
- **The Visuals:** I built a scatterplot mapping **Total Passes (X-Axis)** vs **Total Progressive Passes (Y-Axis)** for whole teams. I highlighted Spain, Argentina, and France. Act 2 featured a horizontal pitch map of Lionel Messi's progressive passes, and Act 3 showed a side-by-side sonar comparison of Aurélien Tchouaméni (safe) and Luka Modrić (progressive).
- **Why it Failed:** Using *absolute totals* was a massive data visualization error. Argentina and France played 7 matches, while Spain only played 4. Of course Argentina had more total progressive passes, but they played almost twice as many minutes! The data was heavily skewed by match count. Furthermore, the narrative jumped randomly from team stats, to Messi, to Tchouaméni, without a cohesive thread.

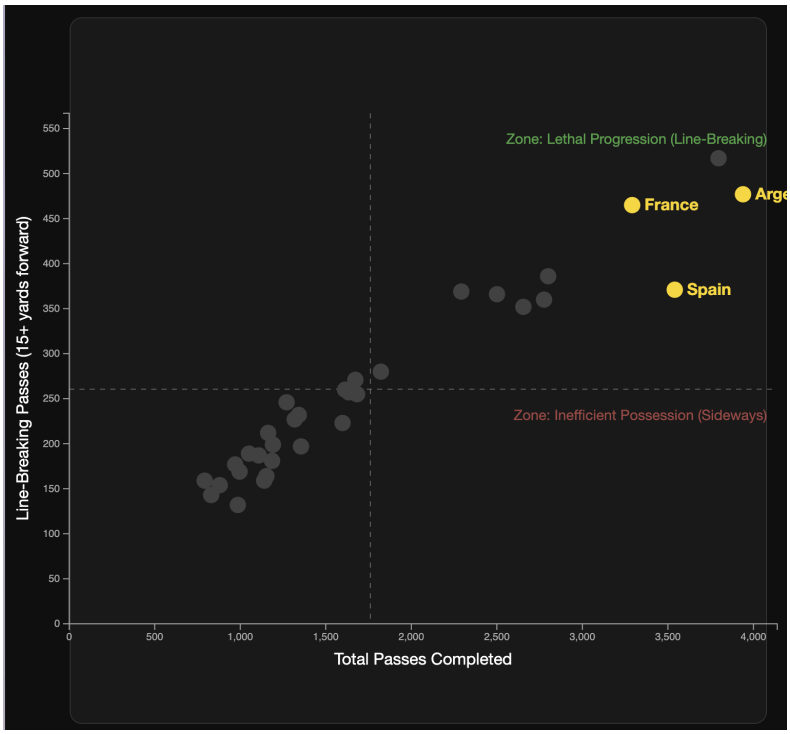


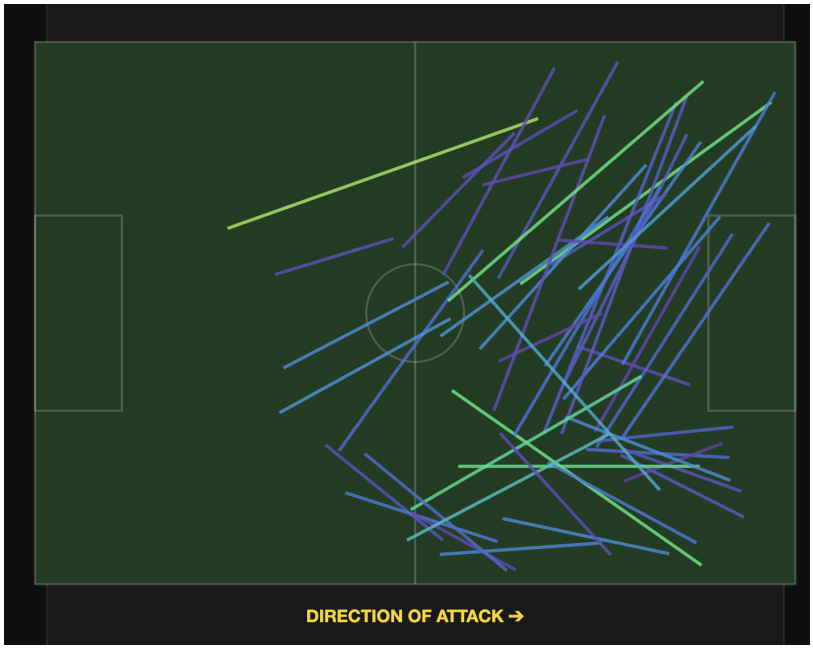

Messi's Line-Breaking Passes


Tchouaméni (Safe)
Modrić (Progressive)

## Attempt 2: Adding Visual Context

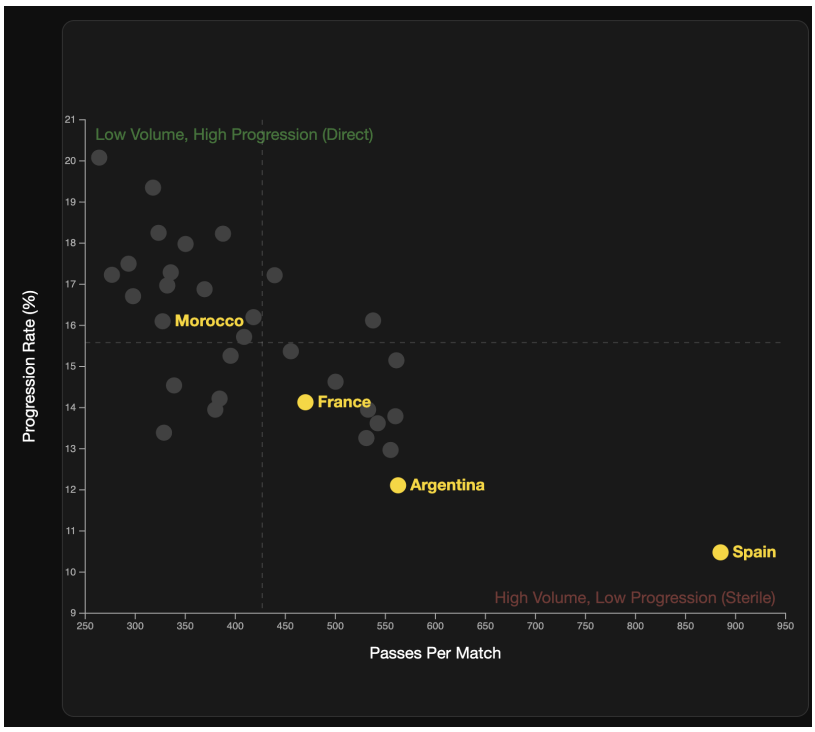
- **The Concept:** I recognized the scatterplot was hard to read, so I tried to fix it with visual design rather than fixing the underlying data.
- **The Visuals:** I kept the flawed "Total vs. Total" axes, but added tournament averages as dotted crosshairs to create four quadrants. I manually labeled the bottom-right as "Inefficient Possession" and the top-left as "Lethal Progression." I also upgraded the aesthetics, adding a green grass background to the pitch map and an "Attacking Direction" arrow to the sonars.
- **Why it Failed:** The aesthetic upgrades were solid (and survived to the final project), but the core problem remained: the crosshairs just highlighted the bias of teams who advanced further in the tournament. I needed to fundamentally change the math.

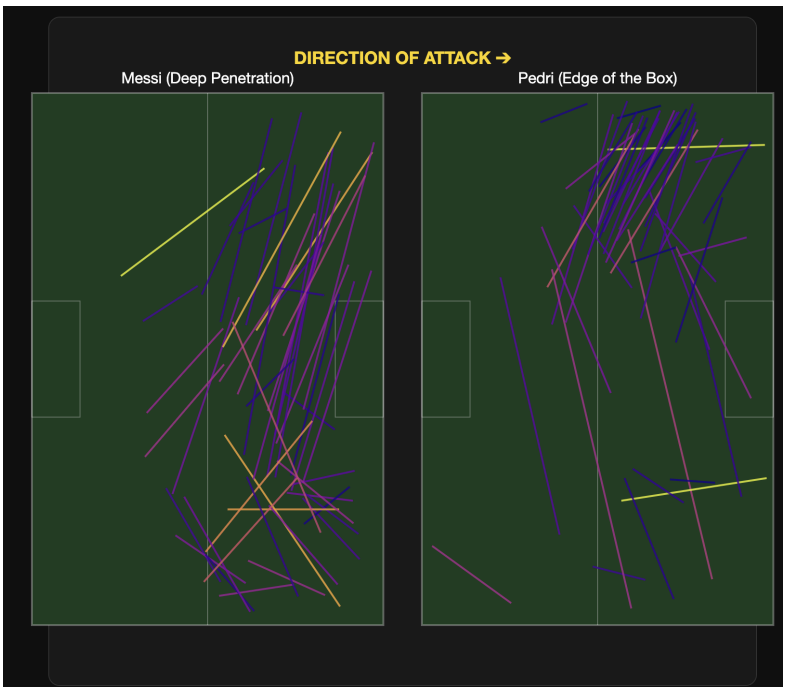


## Attempt 3: The Shift to Ratios & Direct Comparisons

- **The Concept:** I realized I needed to normalize the data. I changed the title to "The Myth of Possession" and fundamentally altered the axes.
- **The Visuals:** Act 1 now plotted **Passes Per Match (X-Axis)** vs **Progression Rate % (Y-Axis)**. This mathematically corrected the match-count bias and perfectly exposed Spain as a massive outlier (885 passes a game, but only a 10.4% progression rate). Act 2 introduced a *split* pitch map comparing Messi (passes piercing the box) vs Pedri (passes cycling outside the box). Act 3 focused on a single sonar of Sergio Busquets to visualize "sterile" sideways passing.
- **Why it wasn't the final answer:** The math was finally sound, and the split-screen comparisons were highly effective. However, I realized I was committing a cardinal sin of data journalism: **Cherry-Picking.** I was hand-picking specific players (Messi, Pedri, Busquets) to fit my pre-written script about Spain vs Argentina. I wasn't actually *scouting* the dataset; I was forcing the data into a fixed story.



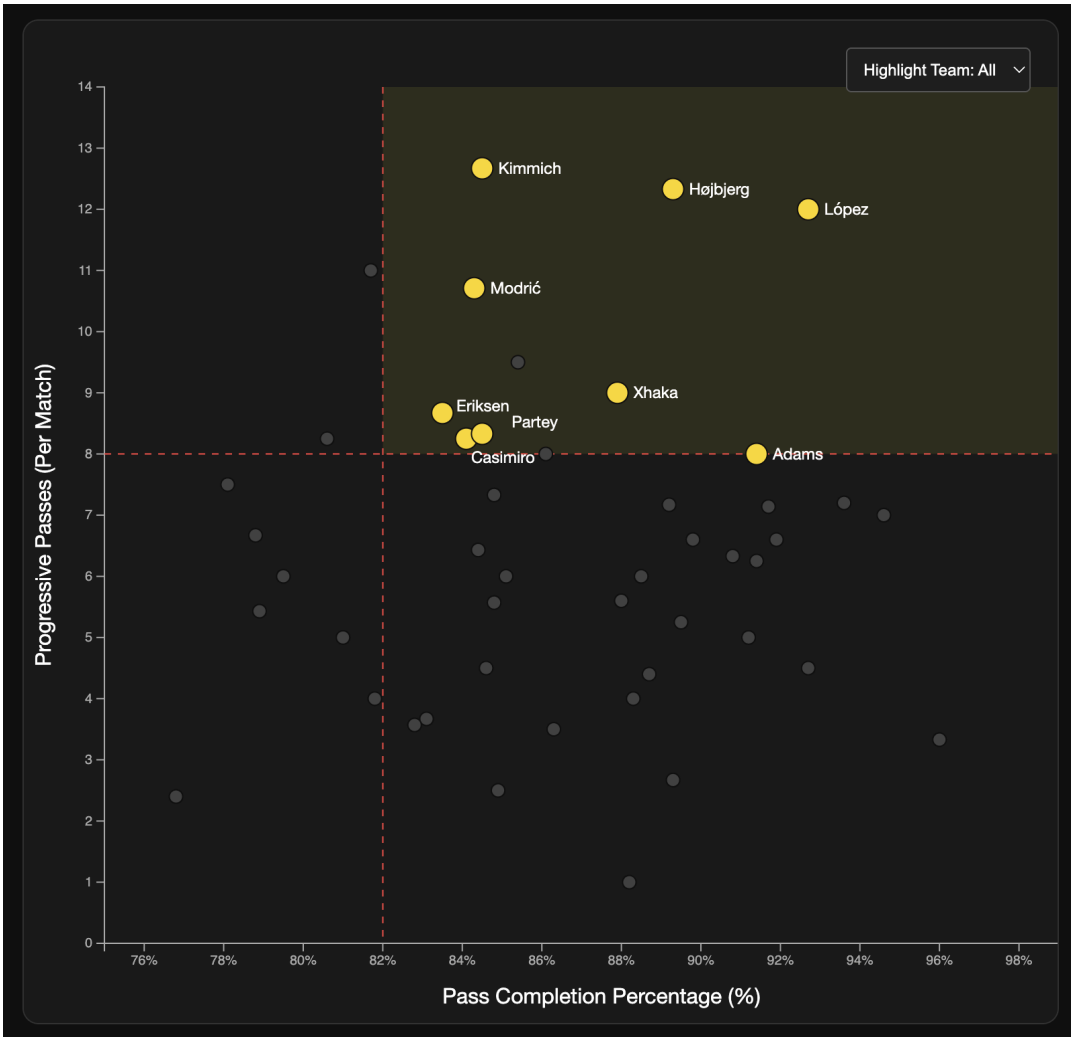



# 4. The Epiphany: The Data-Driven Funnel

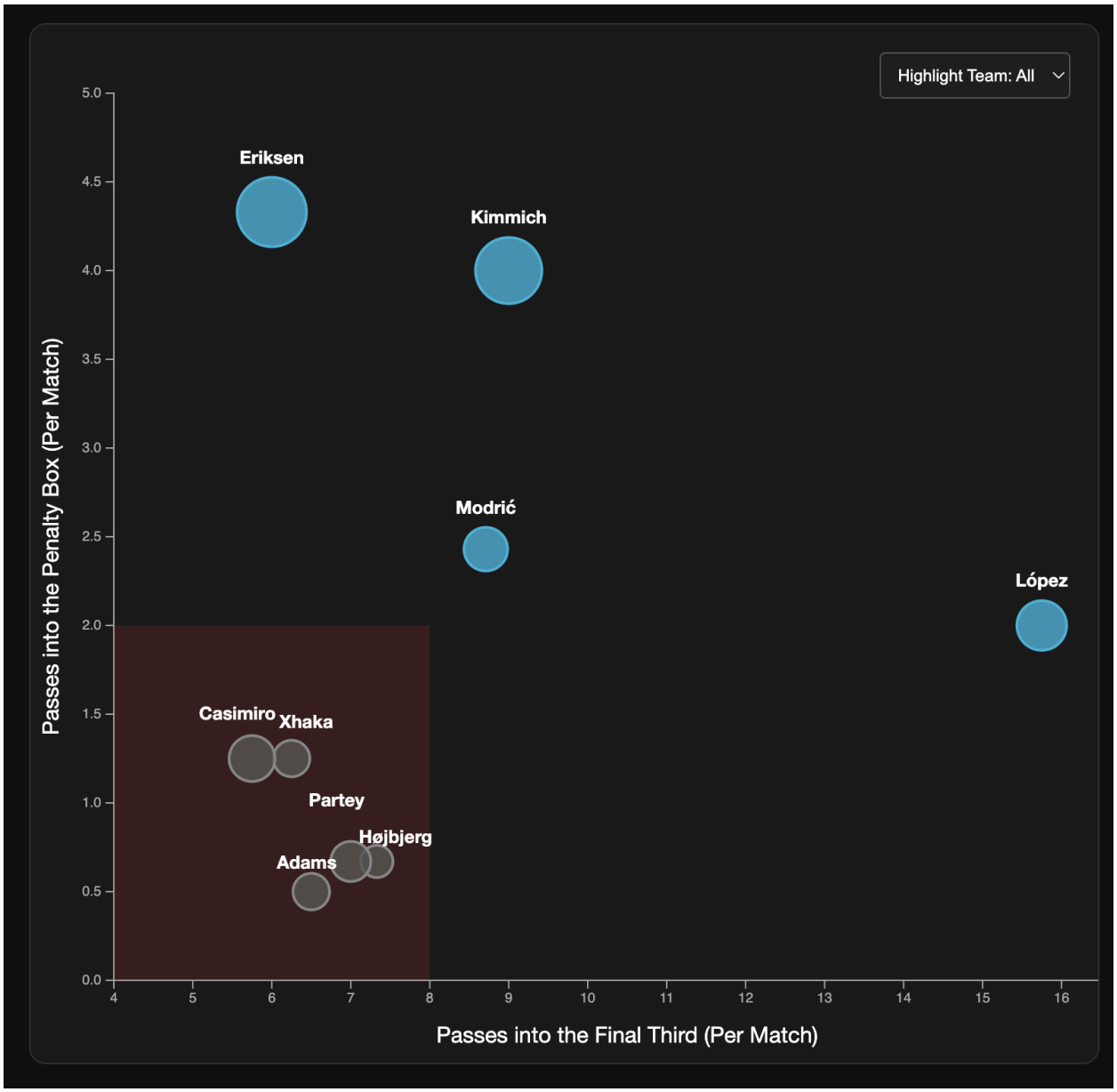
Looking back at my three attempts, I realized that true scouting isn't a static dashboard, it's a **Funnel**. I needed to drop my preconceived notions, process the entire dataset programmatically, and let the data reveal the players to me.

I wrote a series of Python scripts (all_player_data.py, risk_reward.py, funnel.py) to systematically execute this logic:

1. Extract every pass from every midfielder.
2. **Act 1 (The Engine Room):** Filter the 47 midfielders for volume (>45 passes/match), safety (>82% completion), and verticality (>8 progressive passes). This drops the pool to 9 elite players.



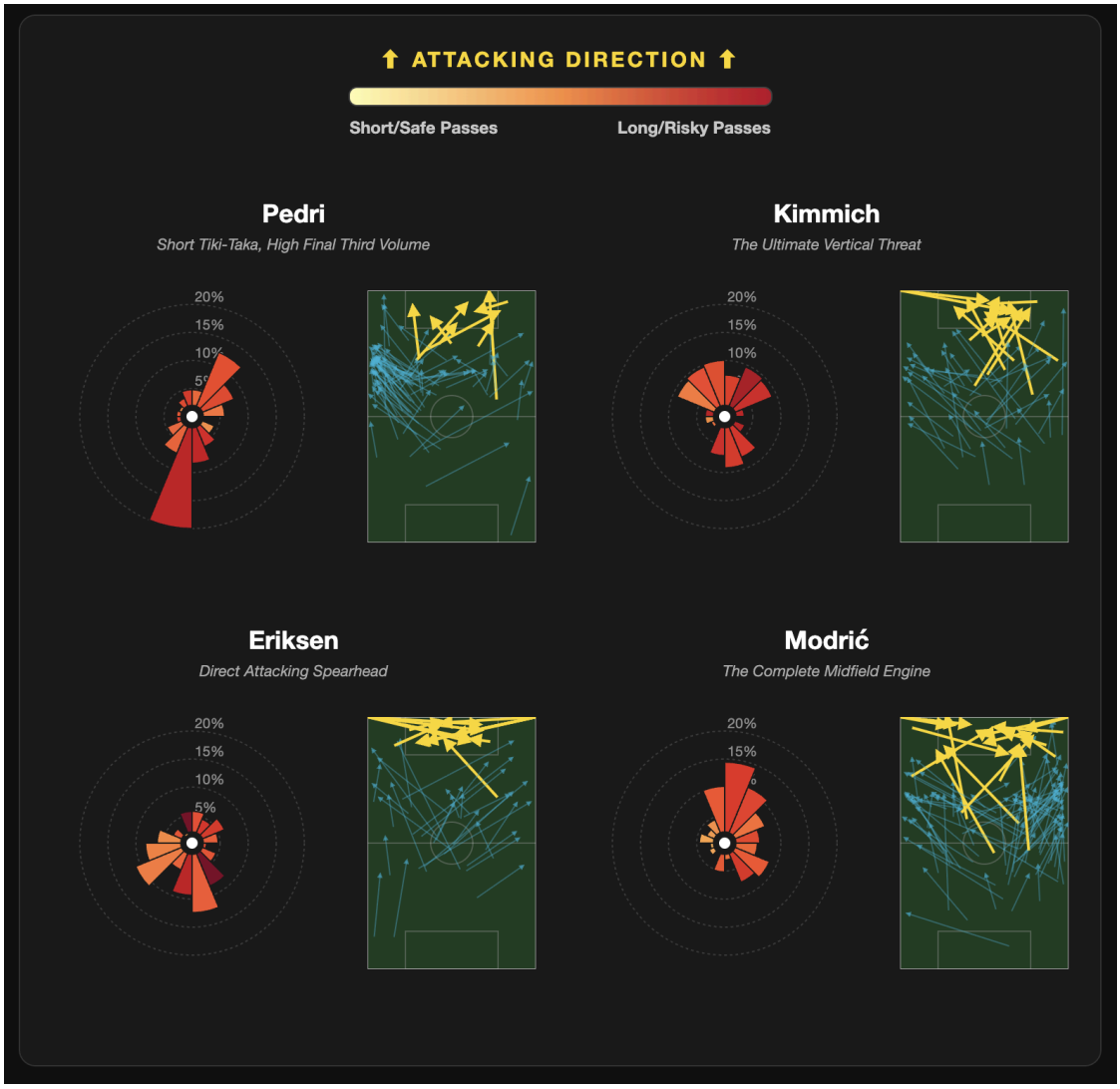
3. **Act 2 (The Final Third Threat):** Zoom in on the Top 9. Map their passes into the Final Third vs. the Penalty Box. Eliminate players who play too deep, leaving us with the true Final 4: Pedri, Eriksen, Kimmich, Modrić.



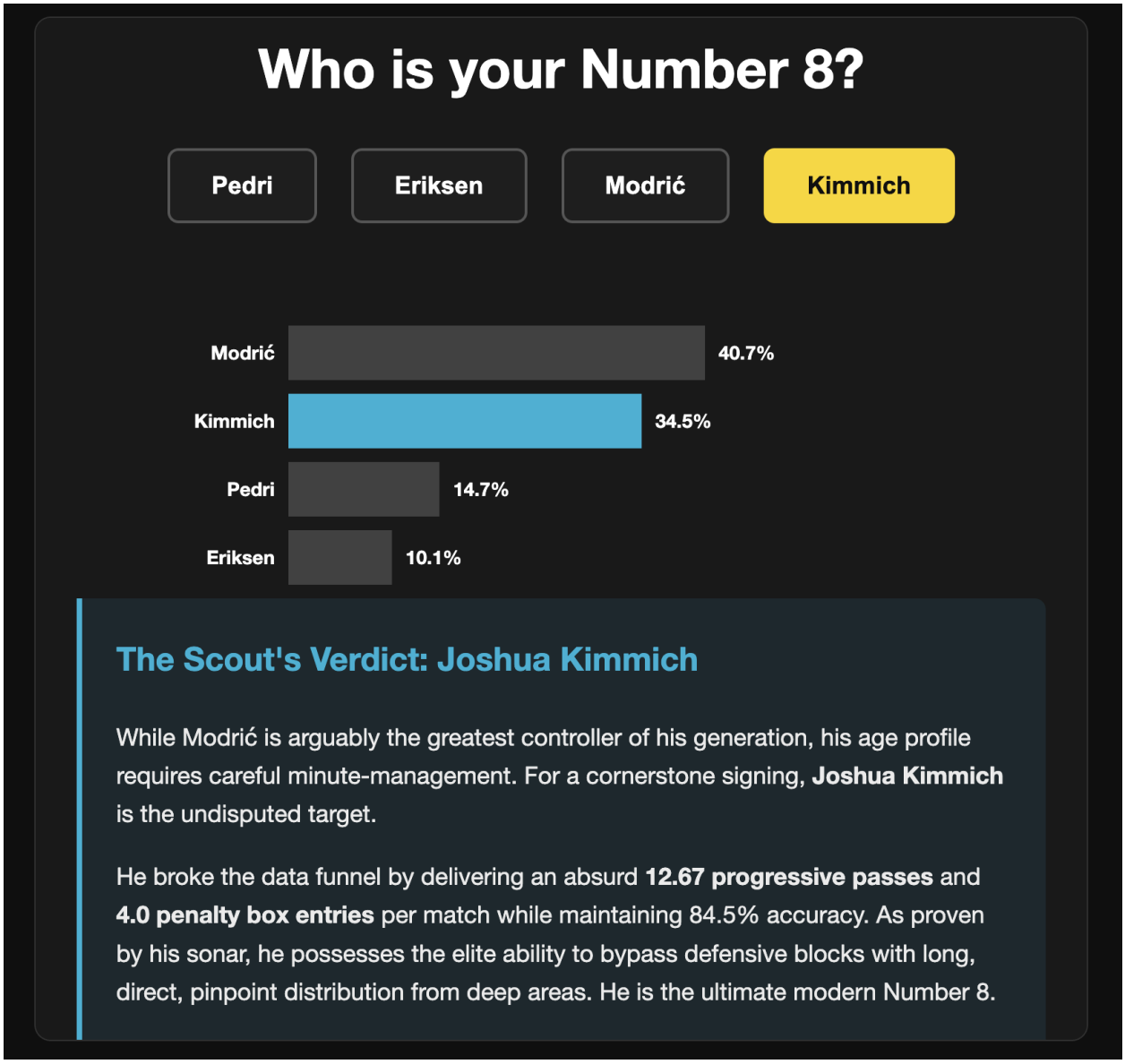
# 5. Visualizing the "How": Sonars and Pitches

Once the funnel mathematically isolated the Final 4, I needed to show *how* they achieved their numbers for Act 3. I wrote sonars.py and pitch_passes.py to extract precise pass angles, distances, and X/Y coordinates.

- **The Sonar Testing Phase:** During development, I briefly considered mapping the *length* of the sonar slices to *pass distance* (yards) and the *color* to *frequency* (%). However, I realized this creates a "Visual Weight Bias." A single 50-yard panic clearance would create a massive visual spike, tricking the eye into thinking the player constantly passes backward. I actively chose to stick to the industry standard: Length = Volume (%), Color = Distance.
- **The Vertical Pitch:** I mathematically mapped StatsBomb's horizontal (120x80) coordinates onto a vertical SVG, making it intuitive to read alongside the upward-facing sonar.



## 6. Conclusion (Act 4)

The final iteration works because it respects the viewer's intelligence. By using Scrollama, the user watches the dots gracefully animate and eliminate themselves based on objective rules.

Finally, I added **Act 4: The Verdict**. Instead of just telling the user who the best player is, I built an interactive voting dashboard. The user is invited to review the data, select their Number 8, and see their vote animate into a community consensus chart before I reveal my final, data-backed conclusion: Joshua Kimmich is the ultimate vertical playmaker. I initially wanted to include a database to collect votes for this part, but with time constraints, I had to hard code some vote data and just simulate it to make it feel like real user votes.